

Python - Dictionary View Objects

The `items()`, `keys()`, and `values()` methods of dict class return view objects. These views are refreshed dynamically whenever any change occurs in the contents of their source `dictionary` object.

The items() Method

The **`items()`** method returns a `dict_items` view object. It contains a `list` of `tuples`, each tuple made up of respective key, value pairs.

Syntax

Following is the syntax of the `items()` method –

```
Obj = dict.items()
```

Return value

The **`items()`** method returns `dict_items` object which is a dynamic view of (key,value) tuples.

Example

In the following example, we first obtain the `dict_items` object with `items()` method and check how it is dynamically updated when the dictionary object is updated.

```
numbers = {10:"Ten", 20:"Twenty", 30:"Thirty",40:"Forty"}
obj = numbers.items()
print ('type of obj: ', type(obj))
```

```
print (obj)
print ("update numbers dictionary")
numbers.update({50:"Fifty"})
print ("View automatically updated")
print (obj)
```

It will produce the following output –

```
type of obj: <class 'dict_items'>
dict_items([(10, 'Ten'), (20, 'Twenty'), (30, 'Thirty'), (40, 'Forty')])
update numbers dictionary
View automatically updated
dict_items([(10, 'Ten'), (20, 'Twenty'), (30, 'Thirty'), (40, 'Forty'), (50, 'Fif
```

The keys() Method

The **keys()** method of dict class returns dict_keys object which is a list of all keys defined in the dictionary. It is a view object, as it gets automatically updated whenever any update action is done on the dictionary object.

Syntax

Following is the syntax of the keys() method –

```
Obj = dict.keys()
```

Return value

The keys() method returns dict_keys object which is a view of keys in the dictionary.

Example

In this example, we are creating a dictionary named "numbers" with integer keys and their corresponding string values. Then, we obtain a view object "obj" of the keys using the keys() method, and retrieve its type and content –



```

numbers = {10:"Ten", 20:"Twenty", 30:"Thirty",40:"Forty"}
obj = numbers.keys()
print ('type of obj: ', type(obj))
print (obj)
print ("update numbers dictionary")
numbers.update({50:"Fifty"})
print ("View automatically updated")
print (obj)

```

It will produce the following **output** –

```

type of obj: <class 'dict_keys'>
dict_keys([10, 20, 30, 40])
update numbers dictionary
View automatically updated
dict_keys([10, 20, 30, 40, 50])

```

The values() Method

The **values()** method returns a view of all the values present in the dictionary. The object is of dict_value type, which gets automatically updated.

Syntax

Following is the syntax of the values() method –

```
Obj = dict.values()
```

Return value

The values() method returns a dict_values view of all the values present in the dictionary.

Example

In the example below, we obtain a view object "obj" of the values using the values() method from the "numbers" dictionary –



```

numbers = {10:"Ten", 20:"Twenty", 30:"Thirty",40:"Forty"}
obj = numbers.values()
print ('type of obj: ', type(obj))
print (obj)
print ("update numbers dictionary")
numbers.update({50:"Fifty"})
print ("View automatically updated")
print (obj)

```

It will produce the following **output** –

```

type of obj: <class 'dict_values'>
dict_values(['Ten', 'Twenty', 'Thirty', 'Forty'])
update numbers dictionary
View automatically updated
dict_values(['Ten', 'Twenty', 'Thirty', 'Forty', 'Fifty'])

```

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